## *Part A*

**Question 1:**

Let

$$\alpha = \sqrt{13} + \sqrt{n}$$

**a)**

$$\begin{aligned}
\alpha^3 - (n+39)\alpha &= (\sqrt{13} + \sqrt{n})^3 - (n+39)(\sqrt{13} + \sqrt{n}) \\
&= (\sqrt{13} + \sqrt{n})(\sqrt{13} + \sqrt{n})^2 - (n+39)(\sqrt{13} + \sqrt{n}) \\
&= (\sqrt{13} + \sqrt{n})(13 + n + 2\sqrt{13}\sqrt{n}) - (n+39)(\sqrt{13} + \sqrt{n}) \\
&= (\sqrt{13} + \sqrt{n})((13 + n + 2\sqrt{13n}) - (n+39)) \\
&= (\sqrt{13} + \sqrt{n})(-26 + 2\sqrt{13n}) \\
&= -26\sqrt{13} + 2 \cdot \sqrt{13}\sqrt{13n} - 26\sqrt{n} + 2\sqrt{13n^2} \\
&= -26\sqrt{13} + 2 \cdot 13\sqrt{n} - 26\sqrt{n} + 2\sqrt{13}n \\
&= -16\sqrt{13} + 26\sqrt{n} - 26\sqrt{n} + 2\sqrt{13}n \\
&= -26\sqrt{13} + 2\sqrt{13}n \\
&= \sqrt{13}(2n - 26)
\end{aligned}$$

Hence,

$$\alpha^3 - (n+39)\alpha = \sqrt{13}(2n - 26)$$

and we can write this as,

$$\sqrt{13} = \frac{\alpha^3 - (n+39)}{(2n-26)}$$

So

$$\sqrt{13} \in \mathbb{Q}_\alpha$$

and since $\sqrt{n} = \alpha - \sqrt{13}$, it follows that

$$\sqrt{n} \in \mathbb{Q}_\alpha$$

Hence,

$$\mathbb{Q}_\alpha = \mathbb{Q}(\sqrt{13}, \sqrt{n})$$

Hence,

$$\mathbb{Q}(\sqrt{13}, \sqrt{n}) = \mathbb{Q}(\sqrt{13} + \sqrt{n})$$